外　国　語
(英　　語)

前期日程

注 意 事 項

1. 試験開始の合図があるまでは，この問題冊子を開いてはいけません。
2. 問題用紙は19ページで，問題は3問あります。全問に解答しなさい。
   解答は，解答用紙の該当欄に記入しなさい。
3. 解答用紙は3枚(その1～その3)あります。
4. 受験番号を，すべての解答用紙の受験番号欄に正確に記入しなさい。
5. 試験中に問題用紙及び解答用紙の印刷不鮮明，ページの落丁・乱丁，汚れ等に気付いた場合は，手を挙げて監督者に知らせなさい。
6. 試験時間は90分です。
7. 試験終了時に，監督者の指示に従って，すべての解答用紙を提出しなさい。
8. 試験終了後，この問題冊子は持ち帰りなさい。

25-前-英

1 次の英文を読んで、1から15の設問について、A~Dの選択肢からもっとも適切なものを選びなさい。＊のついた語には注がついています。（配点 30）

The surprising strategy behind running the fastest marathon

In 2019, Kenyan long-distance runner Eluid Kipchoge became the first person to run a marathon in under two hours. This achievement had held an almost mythical status—existing, as some physiologists projected, at the edge of a human body's capabilities. But in the years leading up to Kipchoge's feat, he and other elite athletes had squeezed their marathon times ever closer. Finally, four years ago, in an unsanctioned event along a flat six-mile loop in Vienna, Kipchoge cruised at record speed, completing 26.2 miles in one hour, 59 minutes, and 40 seconds.

Officially, the two-hour marathon barrier has not been broken, according to World Athletics, the organization that keeps race records. For one, there were no other competitors in Kipchoge's event. Plus, he wasn't alone. He ran in a pack with several expert runners, known as pacers or "rabbits," creating an aerodynamic shape around him. Wind tunnel tests and computer simulations helped the team finesse the formation: Five pacers positioned in front made a V, like the inverse of a flock of geese, while two additional racers ran slightly behind the marathoner, at his left and right flank. After each loop, the pacers rotated out for a fresh set of legs. With this protection from headwinds, Kipchoge completed the distance more than a minute faster than he ever had.

But as it turns out, there may be an even better drafting formation for running a marathon—or so says a team of researchers at École Centrale in Lyon, France, who recently tested multiple shapes by placing miniature manikins in a wind tunnel. They've proposed an arrangement, somewhat resembling a swordfish's body, that would reduce drag forces by 60 percent on a runner. It would result in marathon times about a minute quicker than the Vienna V, they claim. Using the swordfish shape, it "may be possible to run the fastest marathon ever," the study authors write in a paper recently published in the journal, *Proceedings of the Royal Society A*.

But the scientists who helped develop and verify the buzzed-about 2019 configuration dispute that assertion, citing flaws in the new study's wind tunnel setup and the proportions of the mini model runners used to explore it. "The paper itself shows huge differences between present results and those from previous studies," says Bert Blocken, a professor of civil engineering at KU Leuven in Belgium, who used computer and wind tunnel simulations to analyze drag reduction in the Vienna race. While the *Proceedings* paper indicates that a V-shape reduces drag by 50 percent, Blocken says that past work found it was closer to 85 percent.

The benefits of marathon formations are not in dispute here. Aerodynamics researchers agree that packs offer an advantage to going solo, especially at the pace elite runners travel. The drag force acting on an object is proportional to the object's speed squared, points out Pietro Salizzoni, a professor of fluid mechanics and an author of the new study. In other words, the faster you go, the more extreme headwinds you face. On a leisurely stroll, these disturbances are essentially undetectable. But

at the speed Kipchoge ran at for his record—an average of 4.5-minute miles or 13 miles an hour, which would feel like a sprint to most people—pushing air out of the way becomes a literal drag.

For Vienna, three studies pointed to a V-shaped pack, Blocken says: a UK consultancy company's tests of 110 formations using fluid dynamics simulations; his team's own computer simulations of 15 formations; and wind tunnel experiments involving 10 formations. (Confidentiality clauses from INEOS, the British chemical company that sponsored Kipchoge's race, means those reports have not been published.) "All three detailed previous studies gave the same outcome," Blocken explains: The configuration with a "V-shape in front of the target athlete and two runners behind him" produced the lowest aerodynamic drag.

Salizzoni and his coauthors independently tested eight formation styles, including the INEOS V-shape, by mounting stationary 6.5-inch* manikins in an indoor wind tunnel. Their goal was to measure air resistance proportional to what a marathoner would experience. Ultimately, they also found a benefit to placing two pacers in the back. The force acting on a running target "is the sum of the pressure on the front and on behind," Salizzoni notes—those in the rear help minimize any pressure drops. "You want to control the wake* you are producing," he says, similar to the back wings on a Formula One race car.

Where the new findings differ substantially is in the positions of the pacers out front. Salizzoni's team concluded the most effective was a swordfish-profile shape: a lone pacer, followed by four other pacers forming a skinny diamond four feet behind, and finally the target athlete

less than five feet behind the diamond's rear. The "narrower wedge" in this formation could allow runners to "sort of slice through the air," University of Colorado Boulder physiologist Rodger Kram, who wasn't part of the research team, told *Science News*.

Blocken remains unconvinced—he argues that the team's manikins were inappropriately proportioned. "The model used in the study by the present authors seems to be some sort of small cartoon model that is very different from the geometry of a real human body," he says, referring to the unrealistic chest-belly ratio and sharp edges of the models' poseable joints. Blocken's studies used smooth and solid manikins based on scans of real human marathoners.

Salizzoni counters that their figurines had an "equivalent area and an equivalent form" to an athlete, and that using moveable models helped provide more accurate data. After all, humans in motion don't have their arms and legs fixed in place. This "could still give a realistic result for the single runner," Blocken says, but he points out that the fluid dynamics of formations are much more sensitive to subtle changes. In fact, as the *New Yorker* noted at the time, the configuration for the 2019 race was so precisely tuned that if Kipchoge moved five inches out of place, he would be much more exposed to aerodynamic drag.

It may be some time before a top marathoner puts another pack run to the test. Kipchoge typically only competes in two marathons a year. His second race of 2023 will be in Berlin in September, where he officially set the world record in 2022—without the help of a formation—at 2:01:09.

注

6.5-inch	約 16.5-centimeter
wake	航跡

出典

Guarino, B. (2023, August 24). The surprising strategy behind running the fastest marathon. *Popular Science.* https://www.popsci.com/science/marathon-running-formations/

設問

1. According to the reading, Kipchoge's speed in Vienna was under two hours. However, this does not count as breaking "the two-hour marathon barrier" (line 10). What are the reasons for this?
   A. First, Kipchoge ran on a flat running course, with no hills. Second, he was not competing against anyone.
   B. First, World Athletics was not informed of this event. Second, Kipchoge ran on a flat running course, with no hills.
   C. First, Kipchoge ran as part of a group of skilled runners. Second, World Athletics was not informed of this event.
   D. First, Kipchoge was not competing against anyone. Second, he ran as part of a group of skilled runners.

2. How were the pacers used in Vienna?
   A. The pacers were used to help Eluid Kipchoge set a new official record of running a marathon in under two hours.
   B. The pacers were used to increase headwinds for Eluid Kipchoge. Therefore he was able to run the distance of a marathon in record time.
   C. The pacers were used to create an aerodynamic shape around Eluid Kipchoge. As a result, he was able to achieve a record speed.
   D. The pacers ran with Eluid Kipchoge to keep race records official because there were no other competitors.

3. What does “a fresh set of legs” (line 19) mean, as it is used in the reading?
   A. A runner who is well-rested.
   B. A runner with a new pair of high-quality running shoes.
   C. A runner who is new to long-distance running.
   D. A runner with adequate protection against headwinds.

4. What is the “V” in “Vienna V” (line 27), as it is used in the reading?
   A. The formation made by the pacers in the unsanctioned marathon run by Eluid Kipchoge in 2019.
   B. The name of the location for the unsanctioned marathon run by Eluid Kipchoge in 2019.
   C. The elite pacers’ formation was partially in the shape of the swordfish tail which is similar to an upside-down V. “V” therefore represents the tail of a swordfish.
   D. The elite pacers led to Eluid Kipchoge’s victory. “Vienna V” therefore means victory in Vienna.

5. What did Bert Blocken, a professor of civil engineering from KU Leuven, primarily use to analyze drag reduction in the Vienna race?
   A. Field experiments and observations of nature.
   B. A flock of geese and two additional runners.
   C. Elite pacers who can run long distances.
   D. Computer and wind tunnel simulations.

6. What is one disagreement between Bert Blocken and the team of researchers who published a paper in the *Proceedings of the Royal Society A*?
   A. They disagree about whether simulations can be conducted using a wind tunnel.
   B. They disagree about the best way to time running speed.
   C. They disagree about how well the V-shape leads to a reduction of drag.
   D. They disagree about whether marathon formations have any benefits.

7. Based on the reading, what is the benefit of using pacers for a runner to run faster?
   A. Decreased hydration needs.
   B. Decreased temperature exposure.
   C. Reduced drag force.
   D. Reduced muscle fatigue.

8. One of the authors of the paper published in the *Proceedings of the Royal Society A* is Pietro Salizzoni. Which of the following best describes the research of Salizzoni and his coauthors?
   A. They used small manikins inside a wind tunnel to test eight different marathon formations, including the V-shape used in Vienna.
   B. They used small manikins inside a wind tunnel to test eight different marathon formations, not including the V-shape formation used in Vienna.
   C. They used life-sized manikins inside a wind tunnel to test eight different marathon formations, not including the V-shape formation used in Vienna.
   D. They used life-sized manikins inside a wind tunnel to test eight different marathon formations, including the V-shape used in Vienna.

9. What was the purpose of "placing two pacers in the back" (line 64)?
   A. To increase the weight of the formation.
   B. To provide psychological support to the lead runner.
   C. To simulate a race car.
   D. To minimize pressure drops.

10. Based on the reading, which of the following best describes the "swordfish-profile shape" (line 71)?

A. One pacer runs in front. Four other pacers follow the leader, in a straight line. The target athlete runs behind followed by two other pacers.

B. One pacer runs in front. Four other pacers follow in a narrow diamond shape. The target athlete runs behind followed by two other pacers.

C. Four pacers run in a straight line in front. One other pacer follows these four. The target athlete runs behind followed by two other pacers.

D. Four pacers run in a narrow diamond shape in front. One other pacer follows these four. The target athlete runs behind followed by two other pacers.

11. Based on the reading, what is one reason that Blocken thinks that the methods used in his research were better than those used by Salizzoni and his coauthors?

A. The manikins used in Blocken's simulations were based on the bodies of actual marathon runners and had joints that could be moved.

B. The manikins used in Blocken's simulations were smooth and were based on the bodies of actual marathon runners.

C. The manikins used in Blocken's simulations were smooth and had joints that could be moved.

D. The manikins used in Blocken's simulations contained sharp edges and were solid.

12. In the reading, what does Blocken suggest about the fluid dynamics of marathon formations?
    A. They are highly sensitive to small changes.
    B. They require full-sized manikins.
    C. They require aerodynamic drag.
    D. They are less sensitive to manikin proportions.

13. What does it mean to put something "to the test" (line 94), as it is used in the reading?
    A. It means that an athlete tests his or her own physical stamina.
    B. It means that an athlete tries a new sport to see how well he or she can compete.
    C. It means that an athlete tries something to see how well it works.
    D. It means that an athlete tests something in multiple competitions.

14. Which of the following best summarizes the main idea of the reading?
    A. Fluid dynamics simulations should replicate the original situation as closely as possible, so it should be tested along with field experiments such as the one conducted by Eluid Kipchoge in 2019.
    B. Pacers can help to create an aerodynamic shape around a marathon runner and researchers are researching optimal shapes that reduce drag and improve the speed for runners.
    C. It will be necessary to have pacers that can run as fast as the lead marathoner to replicate the field experiment conducted by Eluid Kipchoge in 2019.
    D. Marathoners typically compete in a minimal number of marathons each year. In addition, pack run tests are time and energy consuming, so it will take time before another pack run is tested.

15. Based on the reading, which of the following generalizations is most likely to be true?
    A. When different researchers disagree about something, they generally are unable to state their reasons for the disagreement.
    B. Even when different researchers agree about something, they generally avoid giving each other any credit.
    C. Even when different researchers agree about something, they generally refuse to accept that they might be wrong.
    D. When different researchers disagree about something, they generally try to give reasons for the disagreement.

2 以下の英文を読み、その内容に合うように日本語の要約中の空欄を埋めなさい。（要約は英文の後にあります。）空欄に入れるべき解答の文字数と解答用紙のマスの数は特に連動していないので、20 字以内で必要な長さを日本語で書きなさい。英数字は 1 マスに 2 文字までを記入すること。

例：UEC ➔ [UE] [C]　123 ➔ [12] [3]

下書き用紙が問題の最後にあります。（配点 30）

This robot can tell when you're about to smile—and smile back

With its hairless silicone skin and blue complexion, Emo the robot looks more like a mechanical re-creation of the Blue Man Group than a regular human. Until it smiles.

In a study published March 27 in *Science Robotics*, researchers detail how they trained Emo to smile in sync with humans. Emo can predict a human smile 839 milliseconds before it happens and smile back.

Right now, in most humanoid robots, there's a noticeable delay before they can smile back at a person, often because the robots are imitating a person's face in real time. "I think a lot of people actually interacting with a social robot for the first time are disappointed by how limited it is," says Chaona Chen, a human-robot interaction researcher at the University of Glasgow in Scotland. "Improving robots' expression in real time is important."

Through synced facial expressions, future iterations of robots could be sources of connection in our loneliness epidemic, says Yuhang Hu, a roboticist at Columbia University who, along with colleagues, created

Emo.

Cameras in the robot's eyes let it detect subtleties in human expressions that it then emulates using 26 actuators underneath its soft, blue face. To train Emo, the researchers first put it in front of a camera for a few hours. Like looking in a mirror would do for humans and their muscles, looking at itself in the camera while researchers ran random motor commands on the actuators helped Emo learn the relationships between activating actuators in its face and the expressions it created. "Then the robot knows, OK, if I want to make a smiley face, I should actuate these 'muscles,'" Hu says.

Next, the researchers played videos of humans making facial expressions. By analyzing nearly 800 videos, Emo could learn what muscle movement indicated which expressions were about to occur. In thousands of further tests with hundreds of other videos, the robot could correctly predict what facial expression a human would make and re-create it in sync with the human more than 70 percent of the time. Beyond smiling, Emo can create expressions that involve raising the eyebrows and frowning, Hu says.

The robot's timely smiles could relieve some of the awkward and eerie feelings that delayed reactions in robots can cause. Emo's blue skin, too, was designed to help it avoid the uncanny valley effect. If people think a robot is supposed to look like a human, "then they will always find some difference or become skeptical," Hu says. Instead, with Emo's rubbery blue face, people can "think about it as a new species. It doesn't have to be a real person."

The robot has no voice right now, but integrating generative AI

chatbot functionalities, like those of ChatGPT, into Emo could create even more apt reactions in the robot. Emo would be able to anticipate facial reactions from words in addition to human muscle movement. Then, the robot could respond verbally, too. First, though, Emo's lips need some work. Current robot mouth movement often relies on the jaw to do all the talking, not the lips. "People immediately lose interest…and it's really weird," Hu says.

Once the robot has more realistic lips and chatbot capabilities, it could be a better companion. Having Emo as company on late nights in the robotics lab would be a welcome addition, Hu says. "Maybe when I'm working at midnight, we can complain to each other about why there's so much work or tell a few jokes," Hu says.

出典

Bradshaw, H. (2024, April 15). This robot can tell when you're about to smile—and smile back. *Science News*. https://www.sciencenews.org/article/robot-predicts-smile-facial-expression

【要約】

最近のロボット研究では、(　①　)を 839 ミリ秒前に予測することで、ヒトと同時に笑みを返すロボットの開発に成功した。まず、研究者は、ロボットをカメラの前に置き、自身の表情をモニターさせながら、研究者が制御する駆動装置の動きとそれによって現れる(　②　)との関係を学習させた。次に、(　③　)を分析させて、顔の筋肉の動きから(　④　)を予測するように訓練した。その結果、ヒトと同期して表情を再現できるようになった。表情が同期することで、ロボットに対してヒトが抱くある種の(　⑤　)がなくなると考えられる。今後(　⑥　)すると、より適切な反応ができるようになるであろう。

2 下書き用紙

注意：答えは必ず解答用紙に書きなさい。

英数字は 1 マスに 2 文字までを記入すること。

例：UEC → UE C　123 → 12 3

①

②

③

④

⑤

⑥

3 次の二つの質問から**一つだけ**選んで、少なくとも二つの理由を挙げて英語で具体的に答えなさい。選んだ質問の番号を解答用紙の[　]の中に書きなさい。下書き用紙が次のページにあります。(配点 40)

1. In Japan, two types of calendars, the Western calendar (西暦) and the Japanese calendar (和暦), are used. Should Japan change to using only the Western calendar? Why or why not?

OR

2. Currently in Japan, children over 12 years of age pay adult fees for train and bus tickets. Should this be raised to 18 years old? Why or why not?

3 下書き用紙

注意：答えは必ず解答用紙に書きなさい。